ERC Consolidator Grant 2027

Research proposal [Part B1]

A New Web: Bridging NFPs for Heterogeneous Societal Requirements

ISLANDS

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Abstract

The World Wide Web scaled to billions of users because of architectural decisions that gave up stable references, versioning, and bidirectional links. The downsides of this trade-off show today, when information conveyance has societal consequences: a fabricated audio clip of a party leader circulated, unverifiable, during Slovakia's 2023 pre-election silence; over half the URLs UK ministers gave Parliament in 1997–2006 no longer resolved by 2007; courts in Europe and beyond have received over 1,200 filings with AI-fabricated citations; and copyright courts in Munich and New York must reconstruct attribution trails the Web never recorded. The Web's architecture deferred, rather than avoided, the cost of coordination, and society now pays it.

ISLANDS extends today's Web with an architecture organized around *non-functional properties* (NFPs). *Islands* are bounded networked information spaces whose members choose the NFPs their context demands and provide them jointly. Within an island, NFPs such as *persistence*, *confidentiality-preserving references*, and *attribution* are restored through *immutable versions*, *secure-multi-party-computation-based resolution*, and *traceable transclusion*. Across islands, we establish interoperability through the open Web's existing low-coordination high-scalability mechanisms, thereby also guaranteeing backwards-compatibility with today's World Wide Web. The scientific core of ISLANDS is a *theory of chosen guarantees* that predicts which NFP bundles an island of a given size, stake, and threat model can hold at what coordination cost. We develop this theory, embody it in a reference architecture, and validate it in legal citation, cross-organizational supply-chain, and online social media use cases, where our team has most experience. Our theory and open reference architecture will give society an extended Web whose guarantees are chosen by individual user communities that require them, and on which courts, parliaments, and markets can rely.

Cross-panel / interdisciplinary explanation

The proposal is computer-systems research and is submitted to **PE6 (Computer Science and Informatics)**. It spans four PE6 descriptors: PE6_10 (Web and information systems) as its core, PE6_2 (distributed systems), PE6_5 (cryptology, security, privacy; the SMPC reference resolution of O3), and PE6_7 (artificial intelligence, intelligent systems, multi-agent systems; the agent affordance layer of O4). We expect the scientific contribution to lie in the composition, which none of these sub-fields produces by itself.

Two validation domains and one objective also engage legal and governance scholarship (**SH2**, descriptor SH2_4, legal studies): we design the legal-citation island with legal practice and the social-media island against the data-access and transparency duties of the Digital Services Act, and the stakeholder/incentive analysis (O5) builds on legal scholarship on architecture as regulation, on RFC 9518's centralization analysis, and on the PI's published work on DSA auditing and transparency regulation. SH2 expertise would sharpen the assessment of O5; the methods and deliverables are PE6, and SH2 is flagged as secondary.

Section a: Part I of the Scientific Proposal

b.1 The problem and the idea

The World Wide Web's original design favored simplicity, decentralization, and low coordination overhead, which enabled explosive adoption but sacrificed stable references, built-in versioning, and native bidirectional linking. This was a deliberate choice: Berners-Lee's 1989 proposal required that the system "allow existing systems to be linked together without requiring any central control or coordination" [1], and Fielding and Taylor later clarified the trade-offs that the resulting architecture yields when defining the Representational State Transfer (REST) architectural style: scalable component interactions, uniform interfaces, independently deployable components, and intermediaries that can be inserted anywhere [2]. These priorities suited early document-sharing systems and enabled the Web's scaling up to billions of users and devices within three decades. We argue that the Web's trade-offs are however too rigid for today's use cases, where publishers require traceable dereferencing, supply chain management requires aggregation over confidential data, courts and parliaments require stable, specific, citations, users require sovereignty over their data, and society requires trusted content and accountable authorship.

Over the past twenty years, we have kept the Web's architecture fundamentally static and such properties lacking; however, platforms have responded: Accounts, persistent storage, search, reputation, payment, and a consistent shared state do not exist between independent Web sites; platforms provide them inside proprietary stores into which they draw the Web's content, users, and interactions [3, 4, 5], and they extend into the rest of the Web through APIs and plug-ins on their own terms [6]. Users then rebuild continuity themselves through reposts, screenshots, and off-platform link hubs [7], yielding unsatisfactory mitigation. Berners-Lee warned in 2010 that social networks were walling off the information their users posted [8], and RFC 9518 describes the general mechanism: functions built on open protocols still centralize through network effects and switching costs [9]. The consequences reach the public sphere: Engagement-optimized ranking amplifies some political content over other content [10], and the EU's Digital Services Act now obliges the largest platforms to assess such systemic risks and to open their data to vetted researchers [11], an obligation that remains hard to operationalize because researchers must specify data whose existence they cannot observe [12]. The Commission's first fine under the Act, €120 million against X in December 2025, concerned a "verified" badge that anyone could buy without an identity check and terms that barred researchers from public data [13]. The Web itself authenticates domains through TLS certificates, but it offers no native way to state verifiably who stands behind a document or an account, or what qualifies them [14]. Even the data that platforms do release differs from what their users see [15, 16]. Press publishers face the same opacity: in October 2026, the Paris Court of Appeal upheld an order that Meta negotiate neighbouring-rights payments with French media in good faith, finding that Meta had supplied only fragmentary, opaque, and unverifiable data on its use of their content [17]. Legal scholarship treats such architectural choices as regulation in their own right [18, 19, 20]. And, currently, generative AI repeats the pattern: answer engines and agentic browsers return answers in place of links and so control access to the Web's content [21]. The properties that high-stakes communities need are thus available today, but each inside one platform and under its control.

Society pays for these omissions: In 2007 the UK government found that more than half of the roughly 4,000 URLs ministers had given in answers to Parliamentary Questions since 1997 no longer resolved, and built a redirect service into its national Web archive in response [22]; one in five scholarly articles cites Web content that has since rotted [23]. Two days before Slovakia's 2023 election, a fabricated recording of a party leader discussing vote-rigging circulated with no provenance that voters could check [24]. From February 2027, batteries sold in the EU (and, soon after, products from clothes to detergents) must carry a digital passport with carbon-footprint and due-diligence data assembled from suppliers along the chain [25]. Law shows the cost most sharply, because there a reference's stability is part of the claim it supports. The EU has given judgments and legislation persistent identifiers through ELI [26, 27], but these name documents, not cited passages or versions; half the URLs cited in U.S. Supreme Court opinions no longer resolve, and Perma.cc archives cited pages one citation at a time [28]. By early 2026, more than 1,200 court decisions worldwide concerned AI-fabricated citations [29], and the High Court of England and Wales warned that citing fabricated authorities may lead to referral to the regulator or to contempt proceedings [30]. Content provenance is being moved into camera hardware [31] and into EU law [32, 11], and the Munich Regional Court held in November 2025 that a language model's memorization of song lyrics infringes copyright [33], a similar question to what *The New York Times v. OpenAI* is litigating in the U.S. [34, 35]. What these cases have in common is that, in each case, coordination cost that the Web avoided was deferred, not eliminated; and society now pays it one incident at a time.

The idea. We propose to extend the Web's architecture to build selected properties into the Web itself, in a way that permits communities that require them to select their own property configuration, i.e., a set of desired

non-functional properties (NFPs). ISLANDS introduces the *island* to capture such configured spaces: *Islands* are bounded networked information spaces, such as a legal-citation community, a supply chain, or a community of public-interest publishers on social media, whose members share stakes high enough to justify the cost of coordination through an ex-ante selected architectural trade-off rather than ex-post patching or litigation. Each island selects the NFPs its stakes demand and provides them jointly within its scope, while islands interoperate with one another and with the open Web through the Web's existing low-coordination mechanisms that permit planetary scaling. Within an island, references can for instance be defined as span-level, version-pinned, bidirectional, and verifiable, similar to how Xanadu specified them [36, 37]; across the boundary, they degrade to ordinary Web links, so that every island remains backwards-compatible with today's Web. Islands can also move away from the Web in the opposite direction: an island of fixed, high-speed installations may bind its interfaces once, at design time, in a binary encoding, much as a linker resolves a program's symbols when it builds the program [37, 38]. Entering an island is an explicit admission step: a client, human or agent, is shown the island's NFP set and accepts the obligations of membership, and this is where the island applies its attribution regime. Through ISLANDS, we further explore NFPs that have not yet been proposed on an architectural level. These include *confidentiality-preserving references* that resolve to a value that members can verify was correctly derived from data they may not read; this generalizes secure multi-party computation (SMPC), which has run in production since 2008 [38, 39]. They also include *purpose binding* which attaches the terms of a disclosure (recipient, purpose, validity window) to the reference, so that a reference is re-resolved when its terms expire [40]. *Attribution* binds a reference to an accountable author or, through anonymous credentials and the pseudonyms the European Digital Identity Wallet must support, to an eligible but unidentified one [41, 42, 43]. *Verifiable credentials* let a reference carry an issuer's attested claim about its source, e.g., that a filing was signed by a member of the bar or that an emissions figure was checked by an accredited verifier [14, 44]. *Capability discovery* shows agents, after each interaction, only the actions currently valid, in the hypermedia style of REST [45] as extended to multi-agent systems [46, 47, 48]; current LLM agents instead receive statically linked tool catalogues [49, 50]. The unifying scientific contribution is a **theory of chosen guarantees**: a formal account of which NFP bundles an island can provide, at which coordination cost, under which threat model, and for which stakeholders. Neither hypertext research nor Web engineering offers this today (b.2).

Figure 1 shows islands with different NFP selections, connected to each other and to the open Web. The theory (O1) predicts which selection an island of a given size, stake, and threat model can sustain.

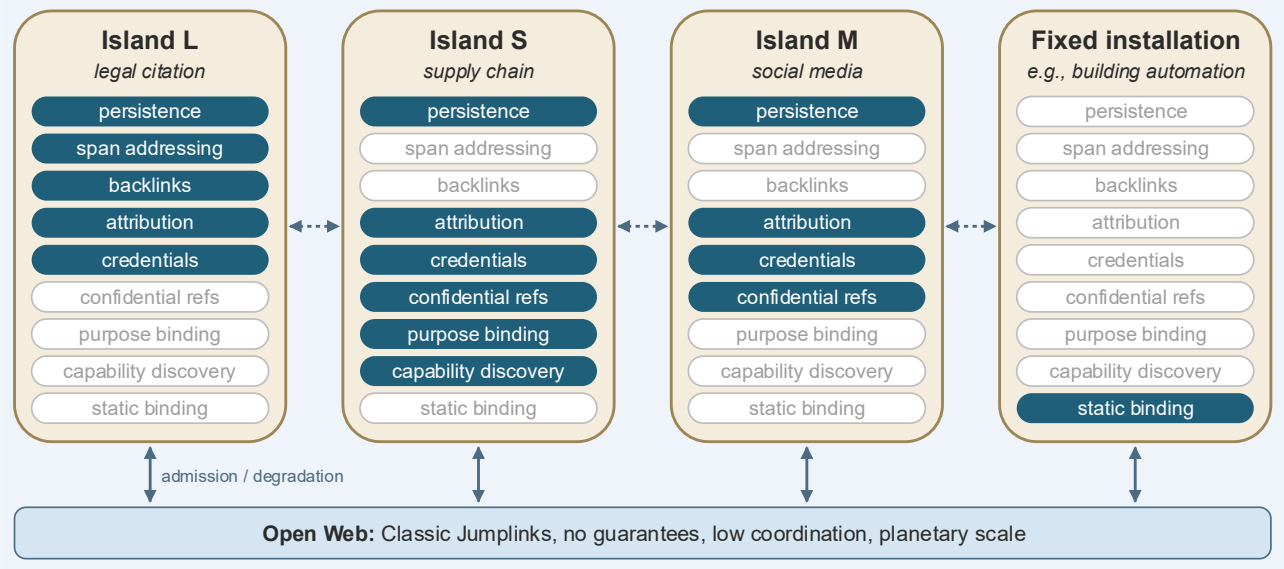


Figure 1. Four islands with different NFP selections (filled: selected). Islands L, S, and M are the validation islands of O5; the fixed installation binds its interfaces at design time (Static Links). Dashed arrows are references between islands; solid arrows are the border crossing, i.e., admission inbound and degradation to Classic Jumplinks outbound. The open Web gives no guarantees and connects all islands at low coordination cost.

b.2 State of knowledge and the gap

Hypertext research assumed rich, versioned, span-anchored links from the start [36, 51, 52], and its agenda included versioning and computed structure [53]. Open-hypermedia systems implemented bidirectional links and referential integrity at workgroup scale [54, 55]; link rot had hence been addressed before the Web made it endemic. The Web dropped these guarantees in order to scale [1, 45]. Remedial infrastructure (e.g., persistent

identifiers, Web archives, Memento [56], Perma.cc [28], and robust hyperlinks [57]) keeps the mutable substrate and builds compensating archives beside it. It thereby neither prevents drift nor verifies integrity, and it reintroduces central coordination points. W3C specifications address passages and backlinks on the open Web: Web Annotation selectors [58] and text fragments [59] identify a passage of a page, and Webmention [60] and Linked Data Notifications [61], which the W3C Social Web Working Group maintains together with ActivityPub [62], deliver opt-in backlinks. None of them pins a version or verifies what it points to. Services assemble bespoke NFP profiles in the same way, e.g., persistence via Perma.cc, provenance via C2PA [31], and identity and notarization via DOIs or blockchains. We read these profiles as evidence of demand for chosen guarantees, but each serves one silo and none can be shared or composed. Content-addressed and versioned storage (Git; IPFS [63]; Software Heritage identifiers [64]) and provenance attestation [31, 65] supply the substrate that Xanadu lacked. They are, however, storage and signing primitives without span semantics, a link layer, or a governance model. Nanopublications come closest [66, 67]. They form a deployed, federated network of immutable, signed semantic statements with self-verifying identifiers, i.e., an island at statement granularity, and may become a component of ours; they do not, however, offer span addressing over arbitrary content, transclusion, or confidentiality-preserving resolution.

Re-decentralization efforts relocate *control* and leave *addressability* as it was. Solid's pods keep location-addressed, mutable resources [68], and data spaces relocate contracts and governance [69]. Federated social media follows the same line: ActivityPub distributes servers but keeps posts location-addressed and mutable [62], and Bluesky's AT Protocol signs and content-addresses each user's repository [70] but leaves references between posts unverified and offers one fixed set of guarantees to all communities. EU data law follows the same line: the Data Act and the portability rights of the GDPR and the DMA move copies of data between holders [71], and the Solid deployments that implement them (cf. our own [72, 73]) leave references location-addressed. Verborgh argues that raw-data transfer cannot sustain a market and proposes trading purpose-bound derivations whose terms limit their shelf life and use [40]. No reference layer can yet name such a derivation, pin it to a version, or carry its terms. Usage control formalized such terms [74, 75], and sticky policies bind them to the data [76]. Enforcement after release, however, requires a trusted monitor at the recipient or falls back on detection after the fact [75]. RFC 9518 explains why open specifications alone do not prevent centralization, and we adopt the design-level mitigations it lists [9]. Network architecture faced the same question one layer below. Clark et al. argued that an Internet embedded in society must leave room for the *tussle* between stakeholders with conflicting interests, and proposed design for change and controlled transparency as its tenets [77, 78]. Plutarch argued for heterogeneous network contexts joined by translating functions at their borders [79]. These positions state principles but predict no outcome, and they concern packets and addresses, not references.

SMPC has run in production since the Danish sugar-beet auction of 2008 [38] and in Estonian government statistics [39], and general-purpose protocols and implementations such as SPDZ [80] and MP-SPDZ [81] make arbitrary computations practical. Our PEP-LCA protocol computes joint supply-chain assessments across up to 50,000 participants without a trusted third party [82], and a follow-on system anchors share disclosure in Solid's access-control layer [83]. Each of these deployments is, however, a bespoke application, and none exposes its result as a reference that others can cite and re-resolve. Information-flow control approached the problem from programming languages. The decentralized label model [84] and Fabric [85] let mutually distrusting principals share persistent objects under confidentiality and integrity labels. They enforce these labels by placing cleartext only on trusted nodes, over mutable, store-identified objects, and they do not quantify what sustaining a policy costs. Verifiable credentials address a further part of the problem: the W3C data model [14], selective-disclosure formats such as SD-JWT [86], and the European Digital Identity Wallet [43] let a holder prove an issuer's claim (e.g., a professional licence) without revealing more than needed [44, 87]. They are, however, presented to one verifier within one session, and they are not bound to the references that the claim concerns. Hypermedia multi-agent systems research argues for run-time affordance discovery [46, 88]. Current LLM-agent practice, in contrast, uses statically linked tool lists [49, 50], which indirect prompt injection turns into an attack surface [89], and no existing design connects capability discovery to a verifiable reference layer.

The gap. We are not aware of any system that composes immutability, span-level verifiable addressing, transclusion, confidentiality-preserving resolution, and agent-facing affordance discovery into one reference architecture. We are also not aware of a theory that predicts which of these guarantees an island of a given size, stake, and threat model can sustain.

b.3 Objectives

The goal of ISLANDS is to find out which guarantees a Web architecture can provide to the communities that need them, and at what cost, and to demonstrate these guarantees in practice. This involves:

- **O1 — Theory of chosen guarantees.** We develop a formal model that relates guarantee bundles (persistence, span addressing, bidirectionality, integrity, confidentiality, attribution, trust, usage, discoverability) to coordination *and economic* costs, island size, threat models, and stakeholder incentives. We instantiate the model as a taxonomy of named **link classes**: a persistence and integrity ladder (Classic Jumplinks, Archived, Trusted, Verified, and Cryptographic Links, in increasing strength) with five cross-cutting axes (attribution, trust, usage, modality, link time). For each class, we prove what it minimally requires, and we derive decision procedures that an island designer can apply.
- **O2 — Reference layer.** We build an addressing and transclusion layer over content-addressed storage. It provides span-level, version-pinned, bidirectional, verifiable references, typed by link class and able to carry purpose-binding terms, and it interoperates with the open Web at island boundaries.
- **O3 — Confidential references.** We turn SMPC-based resolution, which verifies a value without disclosing its inputs, from a bespoke application protocol [82] into a first-class addressing primitive. Each island states its threat model explicitly, from honest-but-curious to malicious-secure.
- **O4 — Agent affordance layer.** We build a capability-exposure layer through which autonomous agents discover the valid next actions in the current state, with link-relation semantics [90] and explicit ordering guarantees for state-changing actions.
- **O5 — Validation islands and incentives.** We instantiate and evaluate islands in three high-stakes domains: legal citation, cross-organizational supply-chain data, and social media. For each class of actor, we analyze whether islands change the incentive to centralize [9].

b.4 Research strategy and methods

To achieve these objectives, we propose a theory work package (WP1) that three systems work packages (WP2–WP4) instantiate and a validation work package (WP5) tests.

WP1 — Foundations: the theory of chosen guarantees (M1–M60, continuous). Method: formal modeling and design-science iteration. We develop a guarantee calculus, i.e., a lattice of guarantee bundles ordered by strength. We formalize the bundles as labels in the information-flow control tradition [84, 91] and extend these labels with persistence and attribution components. Each bundle carries its minimal coordination requirements (e.g., registration, replication, identity, availability coupling) and its admissible threat models. Here we draw on distributed-systems impossibility results [92, 93], simulation-based security definitions [94], and RFC 9518's centralization vocabulary [9]. Some guarantees cannot be enforced once a reference has resolved, but compliance with them can be made incentive-compatible. The calculus hence also scores each bundle by the margin by which compliance is less costly than defection for the actor able to defect. This is the mechanism that transcopyright used for reuse [95] and that current data-economy proposals use for derived insights [40].

To make the calculus checkable, we name each bundle as a *link class* that a reference can declare and a client can enforce, much as link relations [90] or HTTP status codes do at lower stakes. The core is a ladder of persistence and integrity. **Classic Jumplinks** are the Web default (location-addressed, no guarantee) and the fallback outside any island. **Archived Links** stay resolvable through redundancy, as with Memento and Perma.cc [28, 56], but are not verified. **Trusted Links** are registered in the island and tracked bidirectionally, which makes them accountable but not cryptographically checked [55]. **Verified Links** are span-level, version-pinned, and content-addressed, so that resolution matches the cited bytes exactly or fails visibly. **Cryptographic Links** are resolved through SMPC or zero-knowledge proofs (O3) and return a verified derivation from data the client never sees. Further axes cut across the ladder. *Attribution* is non-monotone, since binding actions to identities trades accountability against chilling effects [96, 97]; European courts have weighed this balance since the census judgment [98] and for anonymous online comment [99]. Attribution is hence a pair (Attributed versus Anonymous/Pseudonymous Links [42]), which we evaluate against island governance as well as coordination cost. *Trust* adds Endorsed Links, whose resolution carries a decay-bounded, Sybil-resistant [100] score over retractable endorsements, and **Credentialed Links**, whose resolution returns a verifiable credential about the source together with its current revocation status [14]. Endorsements aggregate the judgment of peers; credentials rest on the authority of an issuer, and the island decides which issuers it accepts. *Usage* adds Purpose-Bound Links, whose terms (recipient, purpose, window) travel with the reference, so that an expired derivation is re-resolved instead of reused. Since no substrate can prevent retention [75], we assess this class by its compliance margin. *Modality* adds Capability Links (O4's state-dependent affordances), to which every rung applies as it does to content. *Link time* records when a reference is resolved, in the sense of program linkers [101]. Capability Links bind as late as possible, at each interaction. **Static Links** bind at design time, in a fixed binary encoding, as Web of Things descriptions already allow for protocols such as Modbus [102]. They give up the evolvability that a uniform interface provides and regain the efficiency it sacrifices [45], which suits fixed, high-speed installations. Like attribution, link time is non-

monotone: the calculus assesses it by when the coordination cost is incurred, once at design time or incrementally at every interaction.

A declared class is a falsifiable claim with two proof obligations. *Realizability* requires that, under its coordination requirements and threat model, a conforming implementation returns the class's guarantee or fails visibly, and never downgrades silently. *Necessity* requires that dropping any one requirement admits an execution that violates the guarantee undetected, i.e., that no rung can be attained at lower coordination cost. For an island of given size, stake, and threat model, the calculus then predicts the highest class it can sustain and at what cost. We develop the calculus iteratively against the WP2–WP4 designs and test it against WP5 measurements: every architectural choice in the systems work packages should be derivable from the theory or force a revision of it. Deliverables are the formal model, the link-class taxonomy, design decision procedures, and a published island design handbook.

WP2 — The reference layer (M1–M42). Method: systems research with a reference implementation. We design the island addressing model. Span-level identifiers are pinned to immutable versions in a Merkle-DAG store, which we build on Git and IPFS primitives [63] and on SWHIDs [64], following RFC 9518's advice to reuse proven mechanisms [9]. A bidirectional link registry scoped to the island makes explicit the coordination that Xanadu wanted globally and the Web refused [36], and transclusion records its provenance [65]. Every reference is typed by its WP1 link class, and the type is enforced structurally. Resolving a Verified or Cryptographic Link thereby returns the exact cited span (or a verified derivation of it) or fails visibly. The registry stores retraction as a superseding record and deletes nothing. A Verified or Trusted Link can hence honour a legal erasure request [103] without breaking the immutable history on which other references depend. Boundary design runs in both directions. Outbound, references degrade gracefully at the island edge: a Verified or Cryptographic Link becomes a Classic Jumplink for a non-member client, carrying a text fragment [59] that keeps the cited passage addressable where the page still contains it, and interoperates with Memento [56] and robust hyperlinks [57]. Inbound, an *admission protocol* (the border crossing) lets a client discover the island's NFP set and bind to it. This is where the island applies its attribution regime (e.g., by asking the client to present verifiable credentials, with selective disclosure [86], or by issuing one), and where the guarantee bundle becomes enforceable on the client as well as on stored content. We benchmark the reference implementation per link class for resolution latency, storage overhead, and link-maintenance cost at island scales of 10^2 to 10^5 members, and we publish all artifacts.

WP3 — Confidential and verifiable references (M6–M54). Method: cryptographic protocol design with proofs and empirical benchmarks. We generalize the PEP-LCA results [82, 83] into a *resolution protocol family*. In this family, a reference's target is secret-shared across island members, and its resolution is a joint computation with a verifiable result. Research arcs: (i) from honest-but-curious to malicious security, using verifiable secret sharing [104] and zero-knowledge correctness proofs, which we expect to yield a quantified privacy-performance frontier; (ii) protocols that tolerate member downtime, which removes the availability coupling we documented in PEP-LCA [82]; and (iii) input integrity through committed inputs and attested sources that connect to hardware-rooted provenance [31]. We prove security in the simulation-based model [94] and evaluate performance on realistic island topologies, together with K. Mitrokotsa and N. Cheng, who co-supervised the PEP-LCA work [82].

WP4 — Agents as hypermedia clients (M12–M54). Method: design and comparative evaluation. We build the island's agent gateway, which exposes capabilities as state-dependent affordances made perceivable through signifiers [105]. The gateway is a hypermedia-driven layer [45] over MCP-class agent protocols [49]. It explains *why* an action is currently valid and gives explicit ordering guarantees for dependent state-changing calls. Each affordance carries the WP1 link class of the resource it exposes, so that an agent can tell a Verified citation from a Classic Jumplink before acting on it. The gateway is also the border crossing for agent clients. An agent's first affordance is the acceptance of the island's NFP set, together with a verifiable credential that states on whose behalf the agent acts, and identity and obligations hence bind before any state-changing call. WP4 extends our work on signifiers [47, 48, 88] towards standardization in the W3C Autonomous Agents on the Web Community Group. We compare task success, safety violations, evolvability, and latency under (a) statically linked tool catalogues, (b) our affordance layer, (c) hybrids, and (d) Static Links over binary protocol bindings, in scenarios drawn from WP5 and from fixed installations in building automation (Siemens) and edge computing (AFarEdge). We expect affordance-scoped exposure to outperform statically linked catalogues on safety and evolvability without loss of task success, and Static Links to outperform both on latency where the installation does not change. The measurements thereby locate, for each scenario, the link time the calculus should recommend.

WP5 — Validation islands and stakeholder incentives (M12–M60). Method: three case studies and a mixed-methods incentive analysis. *Island L (legal citation)* is an end-to-end pilot in which the citations in a working corpus (e.g., a law journal's editorial pipeline or a moot-court record) become span-level, version-

pinned, and verifiable, anchored to ECLI and ELI identifiers where these exist [26, 27]. We measure the citation-integrity failures made structurally impossible, the editorial cost compared to Perma-style manual archiving [28], and practitioner acceptance. Island L also instantiates the *attribution* NFP. It weighs verified authorship, which makes fabrication sanctionable, against pseudonymous regimes, which may shield sensitive filings. Our legal partner is Prof. Tamò-Larrieux (Lausanne), whose group's doctoral work on chilling effects grounds the anonymity side, and Island L docks onto the PI's running law/technology projects (SNSF CoCoDa; *Access to Justice*). *Island S (supply chain)* extends the running PEP-LCA deployment [82, 83] into a full island with confidential references and agent-mediated onboarding, together with its industrial partners (Empa, Siemens). We test it against the disclosure duties of the EU battery passport and the Carbon Border Adjustment Mechanism, which require emissions data from upstream suppliers, including installations outside the EU [25, 106]. Since CBAM requires actual emission values to be checked by accredited verifiers [107], Island S also tests Credentialed Links that carry the verifier's attestation together with the confidential value. *Island M (social media)* gathers public-interest publishers (e.g., public broadcasters, election authorities, and newsrooms) on a federated network [62]. Their posts become version-pinned and attributed, edits and deletions become superseding records, and Credentialed Links state who stands behind an account, which a purchasable badge does not [13]. Vetted researchers resolve Cryptographic Links to exposure statistics that instance operators compute jointly over interaction data that none of them discloses. Island M thereby tests confidential references against the researcher data access of the DSA [11] and against the gap between released and seen data that we documented [15, 16]. We measure the survival of posts and links compared to platform baselines [7], the share of research queries answered without raw-data transfer, and acceptance by publishers and researchers.

The *incentive analysis* studies each class of actor (hyperscaler, legacy industry, OSS/privacy community, browser vendor, platform, public record-keeper) through interviews, institutional analysis of funding models, and the RFC 9518 mitigation checklist [9]. For each class, it asks whether an island changes the actor's incentive to centralize. It also asks whether an island can be configured so that the guarantee-respecting path is the actor's least costly one, since a guarantee that no actor has an incentive to keep is one the theory should decline to promise. The analysis thereby delivers the stakeholder half of O1's decision procedures. It builds on our platform-audit work [15, 16], our working relationship with Solid's steward (the ODI, a CoCoDa partner), and the PI's seat on the European VLOP-auditing oversight board.

Integration and openness. Annual integration milestones bind the WP2–WP4 prototypes into one architecture, which WP5 then evaluates. All software is open source, and all publications are open access. All specifications go to the W3C groups that maintain the protocols islands build on, or to their successors: the Linked Web Storage Working Group (reference layer), the Social Web Working and Community Groups (Island M), the Verifiable Credentials and Federated Identity Working Groups (admission), the Web of Things Working Group (Static Links), and the Autonomous Agents on the Web Community Group (agent layer).

b.5 Team

ISLANDS consolidates the PI's research lines in Web architecture, ubiquitous computing, human-computer interaction, and law. In *Web architecture*, the Web of Things [108], its W3C standardization [102] (now deployed in industrial products), and hypermedia multi-agent systems [46, 47, 88] ground the reference layer (WP2) and the agent layer (WP4). In *ubiquitous computing*, references that bind physical artifacts to digital records [109] and personalization systems that mediate what people perceive in their surroundings [110] extend islands from documents to devices, places, and the people in them. In *human-computer interaction*, the PI's work on affordances and signifiers [48] informs WP4's capability discovery and the admission step at an island's border, where a client is shown the obligations it accepts. In *law*, regulation expressed as technical architecture [20, 111, 112], platform auditing under the DSA [15, 16], and membership in the European VLOP-auditing oversight board ground Islands L and M, the attribution regime, and the incentive analysis (WP5). Each line has so far addressed one part of the problem. The theory of chosen guarantees (O1) needs all of them, because an island's guarantees are at once architectural, experienced by people, and legally consequential. The PI stepped down as Dean in February 2026 to pursue this programme (≥40% commitment).

Two team members carry the running systems on which the project builds. **Luka Bekavac** (through 2029, ≈M20) brings the empirical and regulatory instruments (e.g., platform transparency, DSA data-access auditing, integrity-by-design referencing [15, 16, 109]) for WP5's legal and social-media islands and the incentive analysis. **Jan Grau** (through 2029, ≈M20) built the PEP-LCA SMPC stack and its Solid integration [82, 83], which anchor WP3's first arc and Island S. We will recruit an applied-cryptography postdoc (WP3 lead, from M1), two PhD students (reference layer, WP2, M1; agent layer, WP4, ~M6), a socio-technical postdoc who succeeds Bekavac with overlap (WP5, from ~M17), a further PhD student on the legal island (from ~M21, co-supervised with the legal partner), and a research engineer for the shared prototype. External collaborators are

Mitrokotsa and Cheng (cryptography), Ciortea (hypermedia MAS, USG), Tamò-Larrieux (law, Lausanne), the ODI and the Solid community (via CoCoDa), and the industrial WISER partners (Empa, Siemens).

b.6 Ambition, risk, gain

ISLANDS tests the hypotheses that a coordination cost the Web's designers judged prohibitive [1, 2] is sustainable at island scale, that cryptographic reference resolution can reach interactive latency, and that actors with divergent incentives will adopt shared guarantees. Each hypothesis can be refuted. O1 hence treats guarantees as *chosen*, and O5 measures incentives alongside latency. If our hypotheses hold, we will deliver a validated architecture for the parts of the information space where the Web's founding trade-off now fails at measurable cost, chiefly the parliamentary, legal, and scientific record, public discourse on social media, and regulated industrial data. Autonomous agents could then act on verifiable information in these spaces. If our hypotheses fail, the theory still delimits which guarantees are attainable, for whom, and at what cost.

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Section b: Curriculum Vitae and Track Record

Personal information

Prof. Dr. Simon Mayer; born 16 July 1987; Austrian; three children (2015, 2017, 2019). ORCID 0000-0001-6367-3454.

PhD: Dr.sc. ETH in Computer Science, ETH Zurich; **defense 12 September 2014** (the defense date counts); **eligible** for CoG 2027 under its 5–15-year window (12.3 years on 01.01.2027), no extensions needed. Further degrees: MAS in Secondary and Higher Education, ETH Zurich (2014); B.A. in Economics and Business Administration, University of Zurich (2014); M.Sc. ETH in Computer Science (2010).

Current and previous positions

- 2018–today: **Full Professor of Computer Science** (Interaction- and Communication-based Systems), University of St. Gallen (USG); one of the four founding professors of the School of Computer Science.
- 2024–02/2026: **Dean**, School of Computer Science, University of St. Gallen (stepped down to refocus on research, which frees the time for the ERC ≥40% commitment). 2022–2024 Academic Director; 2018–today Director and 2018–2024 Managing Director, Institute of Computer Science.
- 2020–today: Board Member, **Research Center for Information Law**, USG.
- 2023: Visiting Professor, University of Bologna. 2018–2020: Senior Scientist, Distributed Systems Group, ETH Zurich. 2017–2018: Area Manager "Cognitive Products", Pro2Future GmbH & Project Scientist, Graz University of Technology. 2014–2017: Research Scientist, then Senior Researcher and **Senior Key Expert, Siemens Corporate Technology**, Berkeley, USA. 2015–2017: Visiting Lecturer, UC Berkeley and ETH Zurich. 2013: Visiting Researcher, Ghent University. 2012: Visiting Researcher, MIT.

Ten most significant research outputs

(Chosen to evidence what ISLANDS needs: carrying Web architecture research into standards and products; law/technology co-design; and the two running systems the project builds on.)

1. **W3C Web of Things Thing Description standards**: the PI's PhD and Siemens-era research contributed to the basis of W3C WoT TDs, today deployed in industrial products (e.g., Siemens Desigo building management). Evidence for the exact pathway ISLANDS plans for its island specifications: research results carried into an open standard and from there into practice.
2. Ciortea, **Mayer**, Gandon, Boissier, Ricci, Zimmermann. *A Decade in Hindsight: The Missing Bridge Between Multi-Agent Systems and the World Wide Web.* AAMAS 2019. (Conceptual foundation of WP4; consolidated via two co-organized Dagstuhl seminars, 21072 & 23081, and the PI's founding membership of the W3C Web Agents CG.)
3. Vachtsevanou, Ciortea, **Mayer**, Lemée. *Signifiers as a First-class Abstraction in Hypermedia Multi-Agent Systems.* AAMAS 2023; and Vachtsevanou et al., *Enabling BDI Agents to Reason on a Dynamic Action Repertoire in Hypermedia Environments.* AAMAS 2024. (Run-time affordance exposure; WP4's direct technical basis.)
4. Guitton, Tamò-Larrieux, **Mayer**. ***AI and Law: How Automation is Changing the Law.*** Routledge, 2025. (Book consolidating seven years of the automatically-processable-regulation line; the documented legal-field record the motivation leans on.)
5. Tamò-Larrieux, Guitton, **Mayer**. *A Computational Science Perspective on the Legal System.* **Nature Computational Science**, 2025.
6. **Mayer**, Fuchs, Brügger, Lian, Ciortea. *Improving Customer Decisions in Web-based E-commerce through Guerrilla Modding.* **Nature Machine Intelligence** 3(12), 2021. (User-side re-engineering of Web experiences; an early form of the islands-vs-platform-defaults argument.)
7. Grau, Maftun, Garcia, **Mayer**. *Solid and Secure Multi-Party Computation for a Circular Economy.* Solid Symposium 2026; with Grau's MSc thesis (2025, supervised by the PI with Garcia, Mitrokotsa, Cheng). (Running system behind WP3/Island S.)
8. Bekavac, **Mayer**. *Platforms' Research API Data Access: What Users See vs. What Researchers can Retrieve.* ACM FAccT 2026 (preprint: arXiv:2601.12390); and Bekavac, Strecker-Bischoff, Garcia, **Mayer**, Tamò-Larrieux. *Scrutinizing Systemic Risks in Personalized Recommender Systems Through Sock-Puppet Auditing of VLOPs.* ACM Transactions on Recommender Systems 5(1), 2026. (Empirical basis of the incentive analysis, WP5.)

9. Tamò-Larrieux, **Mayer**, Zihlmann. *Not Hardcoding but Softcoding Data Protection.* Technology and Regulation, 2021; with the *Artificial Intelligence and Law* line on automatically processable regulation (2022–2025). (Law expressed as architecture: the design stance ISLANDS generalizes.)
10. Two CHI 2025 **Honorable Mentions** (adaptive industrial robots; responsible ubiquitous personalization, DIS 2025) and U.S. Patent US11331803B2 (mixed-reality robot programming, with Siemens). (Breadth: HCI venues + transfer to IP.)

Funding record

PI or Co-PI on **4 SNSF, 2 Innosuisse, 3 EU, and 12+ industry/foundation projects** (3.9 MCHF allocated to the PI's group out of 13.6 MCHF total), including: SNSF **HyperAgents** (2019–2024, hypermedia MAS; WP4 ancestry); EU **IntellIoT** RIA (2020–2024); EU **AFarEdge** Chips JU RIA (2026–2029, task leader for the architecture; WP2/WP4 ancestry); Innosuisse Flagship **WISER** (2022–2026, climate intelligence/LCA; Island S ancestry); SNSF **CoCoDa** (2025–2029, VLOP regulation, with Lausanne, Maastricht, and the Open Data Institute; Island L and incentive-analysis ancestry); SNSF/Innosuisse Bridge Discovery **Access to Justice** (2026–2030, with Caritas Switzerland); TA-SWISS *Sprachmodelle und Kompetenzverschiebungen* (2025–2026).

Supervision and team-building

Built a research group and, as founding professor and Dean, a department (10 research groups, 120+ employees, ~300 students, within budget). Five postdocs mentored into faculty/senior positions (Gómez, now Assoc. Prof., TU Braunschweig; Garcia, TT Asst. Prof., IT:U Linz; Ciortea, Asst. Prof., USG; Guitton, Swiss Re; Bektaş, Senior Researcher, KSSG). PhD students: three graduated 2024/25 (Ramanathan, Hostettler, Jha), one completing (Vachtsevanou), six current (including Bekavac and Grau, both on this project's ancestor systems).

Service, standards, recognition

Associate Editor, **PACM IMWUT** (since 2021); Editorial Board, IEEE Pervasive Computing (since 2019); Exec. Editor, Wiley Internet Technology Letters (2017–2023); SI Editor, ACM TOIT (2023). Steering Committee, International Conference on the Internet of Things (2017–); co-organizer, International Workshop on Hypermedia Multi-Agent Systems (2019–2020) and Web of Things workshop series (2012–2017); Dagstuhl seminars 21072 and 23081. Active in **W3C and IETF** standardization; founding member, W3C Web Agents Community Group. Member of the **European oversight board on Very Large Online Platform auditing** (to the PI's knowledge, its only computer scientist). Finalist, Best Teaching Award, USG (2024).

Track-record narrative

ISLANDS consolidates four research lines the PI has built separately: Web architecture, ubiquitous computing, human-computer interaction, and law. Ubiquitous computing (Associate Editor, PACM IMWUT; Steering Committee, International Conference on the Internet of Things) and HCI (two CHI 2025 Honorable Mentions; signifiers carried from HCI to multi-agent systems) connect the two lines described below, carrying Web architecture into physical environments and to the people who act in them. The Web-architecture line built systems that let machines use the Web: from Web of Things architecture (PhD, ETH; Siemens) through its W3C standardization and industrial deployment, to hypermedia multi-agent systems and run-time affordance discovery (AAMAS 2019–2024). This line supplies the architectural competence the reference layer (WP2) and agent layer (WP4) require, with a demonstrated record of moving designs of this kind into open standards and products, and the record is current: the PI presently leads the definition of the architecture in a €27.6M Chips JU action, where a Web-of-Things-based specification authored by the PI's group is the reference that over 1,500 person-months of hardware and platform implementation must satisfy. The legal line studied, with legal scholars, how information control works under regulation: automatically processable regulation, AI-and-law (Routledge 2025; Nature Computational Science 2025), DSA data-access auditing, and service on the European VLOP-auditing oversight board. This line supplies the domain competence Island L and the incentive analysis (WP5) require, and it backs the proposal's legally anchored motivation with the PI's own publications. The two running systems the project extends (the PEP-LCA/ Solid SMPC stack; the signifier-based agent environments) were both built in the PI's group by team members who carry them through the project's first 20 months.